Current Status of Mexican Grey Wolf Population and Recovery Efforts in the Southwestern United States

The MASTERS Program

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BACKGROUND

- Going back to 1915 until 1973, U.S. Fish and Wildlife Service poisoned and trapped almost all Mexican Grey Wolves from the wild.
  - They did it to protect their livestock
  - In 1973, Only 7 Mexican Grey Wolves remained from an original historical population believed to be in the thousands.
- Endangered Species Act was passed 1973
  - Mexican Grey Wolves were added to the endangered list (1976)

PURPOSE

For my research I want to talk about Mexican Grey Wolves and how they actually help us and how we can also help them grow their population to get off the endangered species list.

HOW WE CAN HELP THEM

When we want to help the Mexican Wolf we need to do animal welfare for them. Animal welfare is an animals physical and mental state whether that animal is living or dying.

In order to give animals and the Mexican Wolf the animal welfare they need we need to give them the 5 freedoms (Animal Humane Society).

1) Freedom from Discomfort
2) Freedom to Express Normal Behavior
3) Freedom from Fear
4) Freedom from Great Pain
5) Freedom from Hunger and Thirst

Animals are sentient beings like us (they can have different emotions in different situations), meaning that we need to start giving them similar treatment that we receive.

OTHER OPTIONS TO HELP INCLUDE:

✔Help states and tribes develop strong, conservation-based wolf management programs

✔Advance nonlethal means of preventing conflicts with livestock as standard management and promote their adoption by agencies responsible for managing livestock and wolves

✔Improve the political climate for wolves at state, regional, and federal levels by cultivating grassroots activist and legislators to wolves.

POPULATION

There isn't a "single" factor when it comes to figuring out when a single population isn't considered endangered. Some considerations include:

50/500 Rule
- 50 adults are needed and 500 individuals are needed to avoid long-term extinction

500-5000+ individuals
- to ensure long-term survival in the wild.

(University of Adelaide)

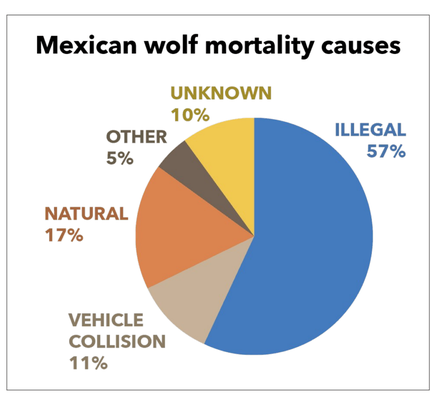

Figure 2: Borrowed pie chart from Project Coyote of causes of Mexican Wolf morality

HOW THEY HELP US

The Mexican gray wolves are very important to our ecosystem. (Lobos of the Southwest)

-Keep pest populations down, which helps stop the spread of disease and parasites.

-Keep elk and deer populations down. It keeps the herds healthy and strong.

-Bring balance to the food chain

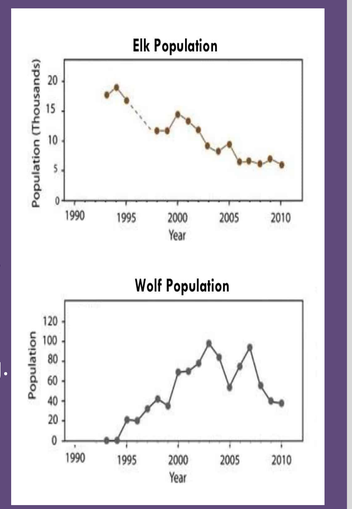

Figure 3: Borrowed graph of elk vs wolf population between 1993-2010

POPULATION OVER THE YEARS

After the Endangered Species Act was passed in 1973 , the population of Mexican Grey Wolves has risen significantly. In this project, data were obtained from US Fish and Wildlife on the recovery efforts in New Mexico and Arizona since 1998, and analyzed in Python using Jupyter Notebooks.

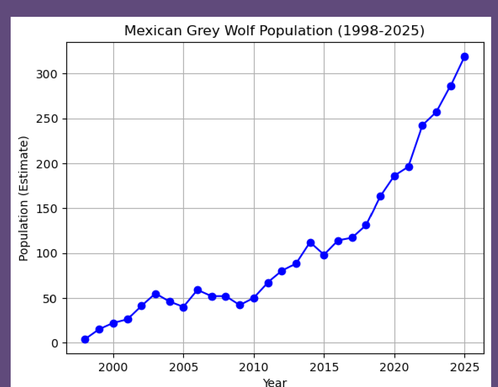

Figure 4: Increase in Mexican Grey Wolf population from 1998-2025 in New Mexico and Arizona.

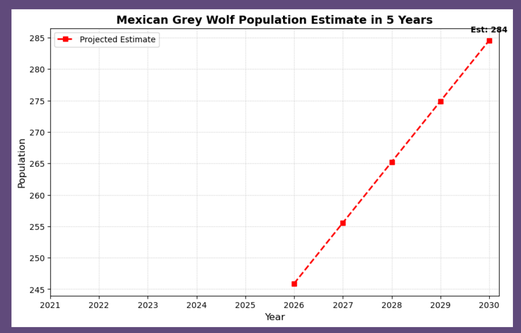

Figure 5: Future estimates of Mexican Grey Wolf population estimate for the next 5 years. Both graphs created by the author using Python in a Jupyter Notebook.

The range of the population that this data was collected from was the Blue Range Recovery Area which is in the Apache National Forest. This are is specialized for the Mexican Grey Wolves to keep them safe from anyone trying to shoot or poison them. (Lobos of the Southwest)

CONCLUSION

We need to help continue the grow of the endangered Mexican Wolves.
There are different solutions but its just a matter of spreading and informing people about how important Mexican Wolves are.

RESOURCES

https://defenders-cci.org/map/Coexisting_with_Mexican_Gray_Wolves/index.html

https://heroftheplanet.org/wolf-coexistence/explore/yellowstone-without-wolves/

https://mexicanwolves.org/in-the-news-agf-reports-48-wolves-in-blue-range-recovery-area/

https://projectcoyote.org/act/learning-hub/mexican-gray-wolf-profile/

https://www.animalhumanesociety.org/health/five-freedoms-animals

https://www.biologicaldiversity.org/species/mammals/Mexican_gray_wolf/index.html

https://www.fws.gov/sites/default/files/documents/Mexican-Wolf-Population-Statisitics_Population-Estimate.pdf

https://www.sciencedaily.com/releases/2009/10/091013104344.htm

https://www.youtube.com/watch?v=ysa5OBhXz-Q